# NAVIS: BUDGET-ADAPTIVE SEMANTIC VIDEO DELIVERY OVER NB-IOT

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## ABSTRACT

Distributed cameras increasingly support surveillance, transportation, and industrial monitoring by offloading visual inference to cloud servers, making uplink delivery a first-order bottleneck. The challenge is acute over Narrowband Internet of Things (such as NB-IoT): it offers broad low-power coverage, but its application-layer service is far below conventional video bitrates and varies over time. We present NAVIS, a feedback-driven semantic video delivery system organized around a **Represent–Adapt–Recover** loop. NAVIS factorizes video into schedulable metadata, reusable background state, and multi-quality regions of interest (ROIs); receiver-confirmed feedback adapts the admission budget online; and persistent receiver state preserves task-relevant visual evidence when current payloads are unavailable. Across the complete MOT17 training set, NAVIS achieves 88.10% pooled ROI availability, compared with 10.10% for ELIC+JPEG baseline at the same 20 kB/round condition. A real NB-IoT run and a multi-node prototype validate end-to-end semantic delivery and cloud-side visual inference.

***Index Terms*—** semantic video delivery, NB-IoT, cloud visual analytics, budget adaptation, region of interest

## 1. INTRODUCTION

Remote visual monitoring increasingly appears in infrastructure-limited settings such as farms, construction sites, and pipelines, where wired backhaul or high-throughput wireless access can be costly, power-hungry, or unavailable. NB-IoT is attractive because it provides wide-area cellular coverage with low device complexity and power consumption [1, 2]; however, its narrow radio bandwidth and time-varying application-layer service make continuous video delivery fundamentally difficult. Such cameras need a visual-delivery abstraction designed for scarce service, not merely a lower-bitrate conventional stream.

Conventional and learned codecs reduce frame-level rate–distortion cost [3–7], while analytics systems adapt resolution, frame rate, ROI quality, or filtering [8–12]. Recent semantic communication further transmits task-oriented representations and adapts them to channel or buffer state [13–17]. Yet even a compressed visual representation may exceed the bytes served in a short NB-IoT interval, while metadata, foreground appearance, and background have different lifetimes. This motivates deciding not only *how* to compress a frame, but *which semantic evidence must be delivered now, which can be deferred, and which can be reused*.

We present **NAVIS**, a feedback-driven semantic video delivery system built around a **Represent–Adapt–Recover** loop. NAVIS factorizes a scene into independently schedulable metadata, reusable background, and track-associated multi-quality ROIs; encodes foreground and background with DCVC-RT; converts receiver-confirmed feedback into an online byte budget; and reuses confirmed cloud-side state when current ROIs are omitted. NAVIS operates above the codec, changing the transmission unit from frames to persistent semantic evidence without modifying the NB-IoT radio scheduler.

The main contributions are:

- **Semantic representation for narrowband visual delivery.** We expose metadata, reusable background, and track-associated multi-quality ROIs as independently schedulable evidence rather than indivisible frames.
- **Feedback-driven evidence allocation.** ACK-confirmed service sets an online byte budget, while semantic value-per-cost ranking decides how much and which evidence to admit.
- **Stateful recovery and end-to-end validation.** We reuse confirmed background and track state when current updates are missing, and validate NAVIS on complete MOT17, a real NB-IoT link, and a two-node prototype.

## 2. RELATED WORK

**Video compression and analytics.** Conventional codecs such as HEVC and VVC exploit spatial and temporal redundancy to optimize rate–distortion performance [3, 4]. Learned video coding has progressed from optical-flow and scale-space prediction to multi-reference, hierarchical, feature-space, and flexible-rate designs [5–7, 18–22]. These techniques reduce bitrate, but their output remains a coded frame/video representation. NAVIS is complementary: it changes the application transmission unit to independently admissible semantic evidence.

Video analytics systems reduce network and compute cost by adapting what is captured, encoded, or processed. ROI coding, AWStream, Reducto, and AccMPEG adjust region quality, source configuration, frame selection, or encoding according to resource and task demands [8–11]. NoScope, Chameleon, Focus, VideoEdge, and Ekya further show the value of specialization and dynamic resource allocation in large-scale video analytics [12, 23–27]. NAVIS extends this idea to rounds too small for a selected visual unit by exposing metadata, ROI qualities, and background refreshes as separately admissible messages.

**Semantic and link-adaptive transmission.** Semantic communication preserves task-relevant meaning rather than every source symbol [28, 29]. Learned joint source-channel coding has also shown that neural representations can adapt directly to noisy wireless channels and feedback [30, 31]. Video systems transmit learned features, keypoints, or semantic increments and adapt bitrate or representations to channel, buffer, or scene conditions [13–17]. NAVIS differs in *where* adaptation acts: instead of mainly adjusting one representation's rate or configuration, it dynamically determines each round's

admissible bytes, evidence composition, ROI quality, and reusable receiver state. This forms a closed-loop semantic evidence allocation architecture across scheduling and recovery.

**NB-IoT communication.** NB-IoT was standardized for low-power wide-area connectivity using a 180-kHz carrier, with coverage enhancement, low device complexity, and cellular scheduling [1, 32]. Studies of its access procedures, uplink adaptation, energy efficiency, coverage, and LPWAN positioning show that delivered service depends strongly on protocol configuration and radio conditions [2, 33–38]. Consequently, an application cannot assume a stable per-camera bitrate even when the physical-layer peak rate appears sufficient for small IoT messages. NAVIS treats the cellular network as a variable service process and closes the control loop using receiver-confirmed application bytes, requiring neither radio-scheduler modification nor privileged base-station information.

## 3. PRELIMINARY STUDY

### 3.1. Compressed-Video Demand versus NB-IoT Service

NB-IoT occupies a 180-kHz carrier and targets low-rate wide-area IoT traffic. Its nominal peak uplink is approximately 250 kbps with multi-tone transmission, while single-tone uplink can be around 20 kbps [1]. These radio-level peaks do not guarantee application service because scheduling, repetitions, overhead, retransmissions, and competing traffic reduce useful payload rate.

The mismatch remains substantial even after modern compression. DCVC-RT compares practical H.265/HM, H.266/VTM, and neural coding in low-rate regimes and reports 21% average bitrate saving over H.266/VTM [7]. For coding rate $r$ bpp, resolution $W\times H$, and frame rate $F$, bitrate is $R = rWHF$. At 1080p30, 0.005 and 0.01 bpp correspond to about 311 and 622 kbps, respectively–already comparable to or above the nominal multi-tone peak and far above single-tone operation before application overhead. Thus stronger compression reduces but does not remove the mismatch. NAVIS therefore encodes foreground and background separately with DCVC-RT and schedules the resulting evidence under the delivered byte budget; Sec. V-D evaluates endpoint-observed service.

### 3.2. From Frames to Semantic Evidence

A compressed frame can remain too large for one service interval, while receiver-confirmed service varies enough that a fixed budget either over-admits in poor periods or wastes favorable capacity. The application should therefore expose fine-grained units whose semantic utility and byte cost can be compared before transmission.

Monitoring scenes naturally contain information with different lifetimes. We represent frame $t$ as

$$\mathcal{V}_t = \{M_t, B_t, \mathcal{R}_t^1, \ldots, \mathcal{R}_t^K\}, \tag{1}$$

where $M_t$ is compact object metadata, $B_t$ is reusable background, and $\mathcal{R}_t^k$ is the $k$-th object ROI. Metadata is frequent and cheap, ROI appearance is updated according to available budget, and background can be refreshed at a slower timescale. Metadata preserves scene structure, ROI payloads restore object appearance, and background updates can be deferred while receiver state remains valid. This hierarchy creates schedulable evidence rather than another image codec.

NAVIS measures service at the receiver. For round $t$, confirmed goodput is $G_t = 8A_t/\Delta t_t$, where $A_t$ is uniquely acknowledged application payload and $\Delta t_t$ is active send-and-ACK time. Fig. 1 shows that the resulting operating quota moves above and below a fixed 20 kB/round reference, motivating feedback-driven admission rather than a preset bitrate.

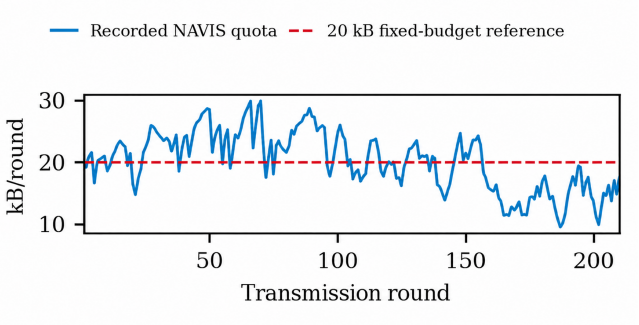



**Fig. 1**. Receiver-feedback-driven quota on MOT17-04 versus a fixed 20 kB/round reference.

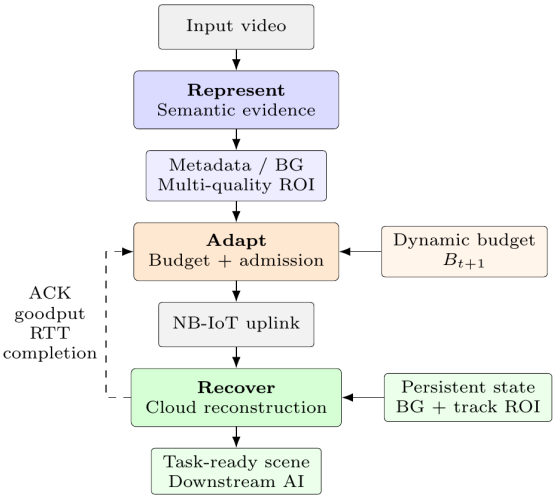



**Fig. 2**. NAVIS Represent–Adapt–Recover architecture.

## 4. SYSTEM

The preliminary study shows that frame-sized transmission units are too coarse for narrowband service, while the number of bytes that can be completed in a round varies over time and cannot be known a priori. Moreover, omitting a current visual payload should not immediately discard useful evidence that has already been delivered. These observations motivate the three-stage **Represent–Adapt–Recover** architecture in Fig. 2. NAVIS first exposes task-relevant video content as independently schedulable semantic units, then adapts both admitted bytes and candidate priority from receiver-confirmed service, and finally reuses persistent receiver state when current evidence cannot be delivered.

### 4.1. Semantic Representation and Admission

A frame bundles together content with very different urgency and persistence: metadata is small but structurally important, foreground appearance changes quickly, and background can often be reused for many frames. Treating them as one transmission unit therefore wastes flexibility exactly when the service budget is tight. NAVIS

instead factorizes each scheduling round into metadata, background state, and track-associated ROIs. A round contains a dynamically formed window of one to five source frames; metadata records frame index, box, class, track ID, motion, and uncertainty and is always admitted first. Background is divided into mandatory state updates and optional refreshes. Foreground ROIs and background are encoded separately with DCVC-RT [7], while metadata remains compact structured messages. NAVIS then schedules these encoded evidence units under the current byte budget.

A second difficulty is that a single-quality ROI creates an all-or-nothing choice: a useful object may be dropped simply because its preferred patch does not fit the residual budget. NAVIS therefore exposes multiple mutually exclusive DCVC-RT quality candidates per ROI. They share track identity and spatial support but differ in size and quality, enabling detail–byte trade-offs without changing association. Representation thus creates fine-grained evidence units for later adaptation.

### 4.2. Feedback-Driven Budget Adaptation

Once evidence is separable, the next question is how much of it should enter a round. A fixed budget is inevitably mismatched to time-varying NB-IoT service: it over-admits when delivery degrades and leaves capacity unused when service improves. NAVIS therefore controls admission from what the receiver actually confirms, rather than from nominal radio rate.

The receiver reports confirmed goodput $G_t$, logical-message failure rate $l_t$, and ACK RTT $R_t$. NAVIS smooths each observation using

$$\hat{z}_t = (1-\alpha)\hat{z}_{t-1} + \alpha z_t, \quad z \in \{G, l, R\}, \tag{2}$$

with $\alpha = 0.3$, and sets the next-round budget as

$$B_{t+1} = \mathrm{clip}\left(\frac{\eta_t \hat{G}_t T}{8}, B_{\min}, B_{\max}\right), \quad \eta_t = \max(\eta_{\min}, \eta(1-\lambda\hat{l}_t)). \tag{3}$$

The clipping bounds prevent short feedback spikes from creating an unrealistic burst, while the loss-dependent safety factor reduces offered load when recent messages fail to complete. Because $G_t$ counts unique acknowledged application bytes, retransmissions are not mistaken for newly delivered semantic evidence. In this way, feedback determines *how much* evidence the next round can safely admit.

### 4.3. Semantic Selection and Stateful Recovery

A byte budget alone still does not determine which ROI updates deserve those bytes. FIFO ignores task value, while size-only selection can favor cheap but uninformative patches. NAVIS therefore combines semantic importance with estimated delivery cost. For ROI candidate $c$ with encoded size $s_c$ and semantic priority $v_c$ (from motion and uncertainty), it uses

$$\rho_c = \frac{1+v_c}{(8s_c/\hat{G}_t + \hat{R}_t)(1+\hat{l}_t)}. \tag{4}$$

Candidates are considered in descending $\rho_c$ and admitted only if their dependencies and residual budget permit. The order is metadata/mandatory background → ranked ROIs → optional background, and selecting one ROI quality suppresses its alternatives. Motion emphasizes rapidly changing objects, uncertainty gives additional opportunity to less reliable tracks, and the delivery-time term penalizes expensive evidence under weak service. Thus the scheduler decides *which* evidence should occupy the bytes made available by feedback.

Even with adaptive admission and semantic ranking, severe scarcity can still force current ROI omissions. Treating every omission as immediate information loss would ignore temporal persistence in monitoring video. NAVIS therefore keeps background and track-indexed ROI appearance at the cloud. A newly received ROI refreshes the corresponding state; otherwise current metadata may reuse a recent confirmed ROI of the same track. With cache TTL $H = 10$ frames, ROI availability is

$$A_{\mathrm{ROI}} = \frac{N_{\mathrm{current}} + N_{\mathrm{proxy}}}{N_{\mathrm{expected}}}. \tag{5}$$

This mechanism reuses previously delivered evidence rather than synthesizing unseen pixels. We therefore report *Current ROI*, which counts evidence delivered for the present frame, separately from *Available ROI*, which also credits an eligible track-consistent proxy.

### 4.4. Multi-Node Coordination

Per-camera adaptation is not sufficient when several cameras share the same constrained service, because a node with many optional ROIs could consume capacity needed by another node's mandatory scene state. NAVIS therefore separates coordination across two levels. A coordinator first protects node $i$'s mandatory demand $m_{i,t}$ and distributes the residual global budget $B'_t$ according to node health and outstanding demand. Each node then applies the same intra-node semantic ranking. Inter-node control assigns *bytes per camera*, while intra-node control selects *evidence per byte*; mandatory metadata remains protected, and no modification to the NB-IoT radio scheduler is required.

## 5. EVALUATION

### 5.1. Dataset and Setup

We evaluate all seven distinct MOT17 training sequences [39] (02, 04, 05, 09, 10, 11, 13), using each physical video once with YOLOv8s-based extraction. NAVIS encodes foreground and background separately with DCVC-RT. Offline replay shares source videos, candidates, and receiver processing across methods. Real-link tests use SIM7080G NB-IoT modems over TCP and an Alibaba Cloud ECS receiver. ROI availability is the fraction of expected physical ROI groups represented by current or eligible cached evidence; reconstructed-frame YOLO detections are matched to receiver ROI records by class and IoU$\geq 0.5$ for F1. We separately report *Current ROI* and *Available ROI* when recovery is involved.

### 5.2. Baseline Comparison

**ELIC+JPEG baseline.** ELIC+JPEG uses the same detections, tracks, metadata, background updates, and ROI crops as NAVIS. Background is encoded with ELIC [40], while each ROI uses one fixed JPEG quality. It fixes the byte budget, admits ROIs in FIFO order, and disables multi-quality selection, feedback control, semantic ranking, and receiver-state reuse, preserving factorization while removing NAVIS's dynamic mechanisms.

**Budget sweep.** We sweep the fixed baseline from 5 to 30 kB/round. Across complete MOT17, ELIC+JPEG rises from only 6.06% to 15.70% pooled ROI availability as the fixed budget increases, whereas NAVIS reaches 88.10% (Fig. 3). At the matched 20 kB/round condition, the baseline achieves 10.10%. The weak response to a

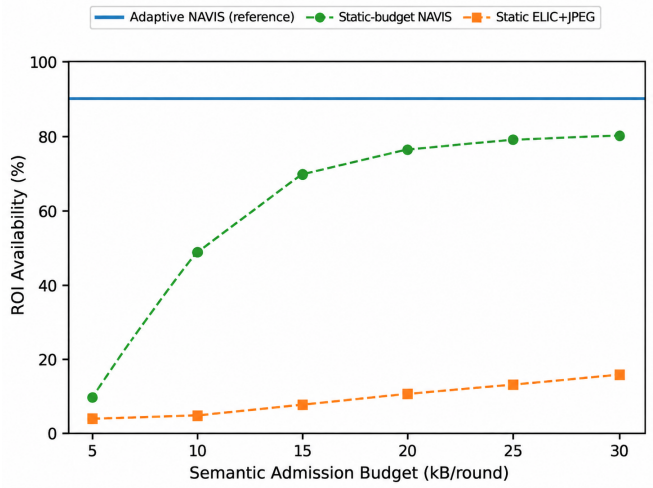



**Fig. 3**. Complete-MOT17 ROI availability versus fixed baseline budgets.

larger fixed budget shows that gains do not come from bytes alone: NAVIS also changes evidence choice, quality, and reuse.

### 5.3. Ablation Experiment

We organize the ablations around the three stages of NAVIS and the admission unit used by the scheduler. The feedback experiment uses a common recorded service trace to isolate budget adaptation.

**Feedback-Driven Budget.** This ablation isolates the **Adapt** stage's control of *how much* evidence enters a round. *Feedback NAVIS* and *Fixed NAVIS* start from 20 kB/round and share the same source windows, candidates, fragmentation, semantic ranking, receiver state, and per-round service trace. Only Feedback NAVIS updates its next-round budget using Eq. (3); Fixed NAVIS remains at 20 kB/round. Their complete-MOT17 ROI availability is 69.53% and 64.16%. This difference comes from matching admitted bytes to the service actually completed by the receiver. When service degrades, feedback avoids over-admitting payloads that are unlikely to finish within the round; when service improves, it can expand the admitted set instead of leaving available capacity unused. This validates adapting admission volume rather than relying on one fixed operating point.

**Multi-Quality ROI Representation.** This ablation isolates the quality flexibility introduced by **Represent**. *Single-Q ROI* keeps the same detections, tracks, ranking, receiver recovery, and per-window budgets, but exposes one quality level per physical ROI. NAVIS instead offers multiple DCVC-RT quality candidates and can admit a cheaper version when the highest-quality patch does not fit. Across complete MOT17, Single-Q achieves 36.95% current and 56.14% available ROI, while NAVIS reaches 68.81% and 88.10%. Lowering an expensive ROI's quality can preserve object appearance while leaving bytes for additional candidates, avoiding an all-or-nothing choice.

**Stateful Recovery.** This ablation isolates **Recover**. The sender, budget, ranking, and transmitted payloads are identical; *No state reuse* only disables track-indexed cached ROI proxies. Both variants therefore deliver the same 68.81% current ROI. Enabling reuse raises available ROI to 88.10%. The improvement comes from exploiting temporal persistence: when current metadata still identifies a tracked object, a recently confirmed appearance can remain useful for a bounded period. Receiver state therefore converts some current-round omissions into usable task context.

**Table 1**. Controlled component ablations on complete MOT17 under constrained transmission budgets.

| Variant | Current ROI | Available ROI |
|---|---|---|
| NAVIS | 68.81% | 88.10% |
| No state reuse | 68.81% | 68.81% |
| Fixed-count admission | 64.37% | 85.93% |
| Single-Q ROI | 36.95% | 56.14% |

**Byte-Level versus Count-Level Admission.** Finally, *Fixed-count admission* retains the remaining NAVIS pipeline but replaces the byte budget with a fixed number of ROI candidates. It obtains 64.37% current and 85.93% available ROI, versus 68.81% and 88.10% for NAVIS. ROI sizes vary with object extent and quality, so equal candidate counts can correspond to very different byte loads. Count-level control can therefore over-admit large ROIs or under-utilize rounds containing cheaper candidates.

**Downstream-task validation.** On archived reconstructions, cloud YOLO reaches 79.06%/77.20% F1 on MOT17-04/09 using class-consistent IoU$\geq$ 0.5 matching, confirming that selective delivery and bounded-age reuse preserve evidence usable by a downstream detector.

### 5.4. Real-Link Validation

On the single-node MOT17-04 run, all 1050 metadata frames are reconstructed. NAVIS provides 4065/4343 available ROIs (93.60%), versus only 820/4343 (18.88%) for ELIC+JPEG. In the two-node prototype, MOT17-04 and 09 are sent in synchronized dynamic windows for 372 rounds. The shared quota ranges from 6–35 kB/round (20.61 kB mean), measured goodput is 5.23 kbps, and p95 ACK RTT is 13.15 s. Both sessions reconstruct every source frame with zero payload decode errors; cloud YOLO reaches 77.58%/78.43% F1. Inter-node quota allocation and intra-node semantic scheduling therefore coexist without cellular-scheduler state.

## 6. CONCLUSION

NAVIS enables cloud visual analytics over low and time-varying NB-IoT service by replacing frame-oriented delivery with dynamic semantic evidence allocation. Its Represent–Adapt–Recover design exposes metadata, reusable background, and multi-quality ROI evidence as independently schedulable units, adapts admitted bytes to receiver-confirmed service, and preserves useful scene context through persistent receiver state when current visual updates cannot be delivered. Controlled ablations show that representation flexibility, feedback-driven admission, byte-aware scheduling, and stateful recovery are complementary. Real single- and multi-node NB-IoT experiments further demonstrate that the same application-layer design remains effective under practical modem scheduling, retransmission, fragmentation, and long acknowledgment delay. NAVIS therefore provides a dynamic delivery architecture for maintaining task-relevant visual evidence when sustained full-frame transmission is impractical.

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